

THE HALF-APÉRY NUMBERS

ABSTRACT. Apéry's $\zeta(3)$ numbers a_n have a square root. Because their generating function is the square of a solution of a second-order operator (the symmetric-square structure underlying Beukers' modular parametrization), the series $u(t) = \sqrt{\sum a_n t^n}$ is a natural object; rescaling by 4^n makes it integral:

$$\tilde{u}_n = 4^n [t^n] u = 1, 10, 534, 40900, 3672550, 360764460, 37546578300, \dots$$

We give the three-term recurrence these numbers satisfy (a half-shifted shape outside the classical Apéry-like boxes), prove their integrality by a two-line universal lemma — and are honest that the integrality is therefore *not* an arithmetic miracle — and then locate the actual miracles: the 2-adic valuation is exactly the binary digit sum, $v_2(\tilde{u}_n) = s_2(n) = v_2\binom{2n}{n}$; and the Lucas congruence $\tilde{u}_{ap+r} \equiv \tilde{u}_a \tilde{u}_r \pmod{p}$ holds *precisely at the primes that split in* $\mathbb{Q}(\sqrt{-6})$ (fourteen primes tested, perfect correlation), the field of discriminant -24 attached to the weight-one modular identity $u(t(q)) = (\eta_2 \eta_3)^{7/2} / (\eta_1 \eta_6)^{5/2}$. The analogous root sequences of the Brown–Zudilin $\zeta(5)$ family share the 2-adic law but have no Lucas law at any tested prime: the split-prime congruence is a fingerprint of the weight-one world, not of root-taking. Every claim carries its evidence label.

1. HOW THE SEQUENCE WAS FOUND

Apéry's numbers $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ satisfy his celebrated recurrence, and their generating function $y_0(t) = \sum a_n t^n$ satisfies a third-order differential operator L . In the course of repairing a proof (at a referee's insistence) it was established exactly that L is the symmetric square of the second-order operator

$$\tilde{D} = \theta^2 - t \left(34\theta^2 + 17\theta + \frac{5}{2} \right) + t^2 \left(\theta + \frac{1}{2} \right)^2, \quad \theta = t \frac{d}{dt},$$

so that $y_0 = u^2$ for the normalized solution u of $\tilde{D}$ [PROVED] (an identity of operators over $\mathbb{Q}(t)$; see the companion expository paper, Lemma 5.1). The series

$$u(t) = \sqrt{y_0(t)} = 1 + \frac{5}{2}t + \frac{267}{8}t^2 + \frac{10225}{16}t^3 + \dots$$

has denominators that are pure powers of 2, and

$$\tilde{u}_n := 4^n [t^n] u$$

is an integer sequence: the *half-Apéry numbers*

$$1, 10, 534, 40900, 3672550, 360764460, 37546578300, 4068324634440, \dots$$

[VERIFIED: exact $\mathbb{Q}$, $n \leq 200$; `eps56_half_apery.py`]. We could not consult the OEIS from this environment; the sequence is displayed prominently so the reader can. A search of the surrounding repository finds no prior appearance.

2. THE RECURRENCE

Reading $\tilde{D}$ coefficientwise and rescaling $t \mapsto t/4$:

Proposition 2.1. [PROVED] (*from $\tilde{D}$; re-verified numerically to $n = 200$*)

$$(n+1)^2 \tilde{u}_{n+1} = (136n^2 + 68n + 10) \tilde{u}_n - 4(2n-1)^2 \tilde{u}_{n-1}, \quad \tilde{u}_0 = 1, \tilde{u}_1 = 10.$$

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Two features are worth savouring. First, $136 = 4 \cdot 34$: Apéry's famous 34 survives, dressed by the rescaling. Second, the trailing coefficient $4(2n-1)^2 = 16(n-\frac{1}{2})^2$ is *half-shifted*: the recurrence lies outside the Zagier and Almkvist–Zudilin normal forms (whose trailing terms are cn^2 or $n(cn^2+d)$), in a half-integer-shifted class of its own — as befits the square root of a weight-two world.

3. INTEGRALITY IS FREE (AND WE SAY SO)

Lemma 3.1 (universal root-integrality). [PROVED] *Let $f \in 1 + t\mathbb{Z}[[t]]$. Then $\sqrt{f(4t)} \in \mathbb{Z}[[t]]$ and $f(8t)^{1/4} \in \mathbb{Z}[[t]]$.*

Proof. $f(4t) = 1 + 4tg(t)$ with $g \in \mathbb{Z}[[t]]$, and $(1+x)^{1/2} = \sum_k \binom{1/2}{k} x^k$ with $4^k \binom{1/2}{k} = (-1)^{k-1} 2 C_{k-1}$ for $k \geq 1$, where C_{k-1} is a Catalan number; so every term of $\sum_k \binom{1/2}{k} (4tg)^k$ is integral. For the quarter root, $v_2 \binom{1/4}{k} = -2k - v_2(k!) = -(3k - s_2(k)) \geq -3k$, so $8^k \binom{1/4}{k} \in \mathbb{Z}$. $\square$

Remark 3.2 (an honest deflation). Lemma 3.1 applies to *any* integral series (a random-series sanity check passes, 5/5). In particular the integrality of $\tilde{u}_n$, and likewise the observations $Q(4t)^{1/2}, Q(8t)^{1/4} \in \mathbb{Z}[[t]]$ for the Brown–Zudilin $\zeta(5)$ row Q_n made earlier in this programme, are *universal*, not arithmetic fingerprints of symmetric-power structure as first advertised. The programme's earlier reading is hereby corrected. The genuine arithmetic lives one layer deeper, in §4 and §5; and the genuinely special fact about *this* square root is not that it is integral but that it is *modular* (§6) — a random integral series' root is neither a solution of a nice operator nor a weight-one form.

The rescaling 4^n is sharp: $u(2t)$ is not integral [VERIFIED: $c^* = 2$].

4. THE 2-ADIC PROFILE: BINARY DIGIT SUMS

Observation 4.1. [VERIFIED: $1 \leq n \leq 200$]

$$v_2(\tilde{u}_n) = s_2(n) = v_2 \binom{2n}{n},$$

the binary digit sum of n (equivalently, by Kummer, the 2-adic valuation of the central binomial coefficient).

The mechanism is visible in the proof of Lemma 3.1: the extreme term of the binomial series contributes $4^n \binom{1/2}{n} = \pm 2C_{n-1}$, and the tidy identity $v_2(2C_{n-1}) = s_2(n)$ [PROVED] (from $s_2(n) = s_2(n-1) - v_2(n) + 1$) makes $s_2(n)$ the generic valuation whenever a_1 is odd; five out of five random odd- a_1 series show the same profile. So Observation 4.1 says the half-Apéry numbers are 2-adically *as generic as possible* — no hidden cancellation — which is itself a statement with content, and it transfers verbatim upstairs: the half- and quarter-roots of the $\zeta(5)$ row obey $v_2 = s_2(n)$ as well [VERIFIED: $n \leq 60$, resp. $n \leq 30$]. We note that central-binomial *divisibility* $\binom{2n}{n} \mid \tilde{u}_n$ fails in general: the coincidence with $v_2 \binom{2n}{n}$ is a 2-adic statement only.

5. THE JEWEL: A SPLIT-PRIME LUCAS LAW

Apéry's numbers satisfy the Lucas congruence $a_{ap+r} \equiv a_a a_r \pmod{p}$ for every prime. Their square root is choosier — and its choice is a quadratic field.

Conjecture 5.1 (split-prime Lucas law). *For an odd prime p ,*

$$\tilde{u}_{ap+r} \equiv \tilde{u}_a \tilde{u}_r \pmod{p} \quad \text{for all } a, r \quad \Longleftrightarrow \quad \left(\frac{-6}{p} \right) = 1 \text{ or } p \in \{2, 3\}.$$

[VERIFIED: fourteen primes $p \leq 47$, all ≈ 2400 cells with $n \leq 200$; perfect correlation] — the law *passes* at $p = 3$ (ramified) and at the split primes 5, 7, 11, 29, 31, and *fails* (massively: over 70% of cells) at the inert primes 13, 17, 19, 23, 37, 41, 43, 47. The prediction was made from the first four primes and confirmed at the next ten.

$\mathbb{Q}(\sqrt{-6})$ has discriminant -24 — exactly the quadratic datum $\prod d^{r_d} = 6$ of the eta-quotient of §6. At inert primes no scalar-twisted law $\tilde{u}_{ap+r} \equiv c^a \tilde{u}_a \tilde{u}_r$ repairs the failure (probed directly); intriguingly, certain rows are constant (e.g. $p = 13$: $a = 2, 5$; $p = 23$: $a = 3$), the shape one expects from a two-term dihedral recursion rather than a one-term twist [OPEN].

The law is *not* an artifact of root-taking: the half-root of the $\zeta(5)$ row satisfies no Lucas congruence at any tested prime (3, 5, 7, 11, 13, all FAIL) despite having the same 2-adic profile. The split-prime law is a fingerprint of the weight-one modular world to which this particular square root belongs.

6. THE MODULAR IDENTITY

Let $t(q) = q(\eta_1\eta_6/\eta_2\eta_3)^{12}$ and $F(q) = (\eta_2\eta_3)^7/(\eta_1\eta_6)^5$ be Beukers' hauptmodul and weight-two form for the Apéry family on $\Gamma_0(6)$ ($\eta_d = \eta(q^d)$).

Proposition 6.1. (1) [VERIFIED: coefficientwise to q^{45} , exact] $u(t(q)) = \sqrt{F(q)} = (\eta_2\eta_3)^{7/2}/(\eta_1\eta_6)^{5/2}$.
 (2) [PROVED] η is nonvanishing on the upper half-plane, so $\sqrt{F}$ is a single-valued holomorphic function there, and its square is a holomorphic weight-two form on $\Gamma_0(6)$: u is a holomorphic weight-one form on $\Gamma_0(6)$ with a multiplier system of order dividing two. The half-integral eta exponents have $\sum_d dr_d = \sum_d (6/d) r_d = 0$, so no fractional power of q appears, consistent with (1).
 (3) [VERIFIED: q^{24}] The q -expansion of u has unbounded 2-power denominators (v_2 of the n -th coefficient $= -(2n - s_2(n))$): the 4^n -integrality of the half-Apéry numbers is a property of the mirror coordinate t , not of the nome. Relatedly, u is not a $\mathbb{Z}$ -linear combination of the discriminant- -24 theta series ($x^2 + 6y^2$, $2x^2 + 3y^2$): checked and false. The $\mathbb{Q}(\sqrt{-6})$ -arithmetic of u surfaces in the Lucas law of §5, not in an integral theta identity.

So the half-Apéry numbers are the mirror-coordinate avatar of a genuine weight-one form — the “square root of Beukers’ parametrization” — and their special congruences see the class field of $\mathbb{Q}(\sqrt{-6})$.

7. UPSTAIRS: THE $\zeta(5)$ ROOT SEQUENCES

For the Brown–Zudilin row Q_n (1, 21, 2989, ...), the same operations give integer sequences

half: 1, 42, 23030, 21898308, 26657064742, ... quarter: 1, 42, 45178, 85695756, ...

[VERIFIED: $n \leq 60$] with $v_2 = s_2(n)$ in both cases, and with no Lucas law at any tested prime. By Lemma 3.1 their integrality is universal; by §5 their failure to Lucas is informative: whatever weight-one-like structure the $\zeta(5)$ five-block’s roots carry, it is not (yet) visible in these coordinates. We record them for the day the five-block’s own uniformization is understood.

8. OPEN PROBLEMS

- P1.** Prove Conjecture 5.1. The natural route: express $\tilde{u}_n \bmod p$ through the weight-one form’s Frobenius structure at p (split primes give two rational unit roots, inert primes a conjugate pair) — the dihedral mechanism that also explains the constant inert rows.
- P2.** Prove $v_2(\tilde{u}_n) = s_2(n)$ for all n (finish the genericity argument for this particular f : no cancellation beyond the extreme binomial term).
- P3.** Identify the inert-prime two-term law suggested by the constant rows.
- P4.** Find a combinatorial or lattice-path interpretation of $\tilde{u}_n$ (the sequence deserves one).

P5. Check the OEIS; if absent, the sequence, its recurrence, and the split-prime law would make a tidy submission.

PROVENANCE

All computations: `work/z5eps/eps56_half_apery.py` with log `eps56.log`; exact `Fraction` arithmetic throughout; the `Sym2` input is Lemma 5.1 of the companion expository paper (`papers_out/expository_a`) proved as an operator identity over $\mathbb{Q}(t)$. Evidence labels: `[PROVED]` = complete proof; `[VERIFIED: range]` = exact computation over the stated range; `[OPEN]` = open. No finite computation is reported as a proof.

REFERENCES

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